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This set of notes combine Josh Pollitz's lecture notes on stable category of Gorenstein rings, [1], and my previous knowledge of derived category (for a reference, see for example [2]).

1 Prerequisites on injective modules

This section summarizes essential properties of injective modules and injective hulls that will be used afterwards.

Definition 1.1. Every R-module M admits an **injective hull** $E_R(M)$, which is the maximal essential extension of M (which ensures existence). It is also the minimal injective module containing M, and any injective containing M has $E_R(M)$ as a direct summand.

Proposition 1.2. *Let p be a prime ideal and $E = E_R(R/p)$. Then $\mathrm{Ass}(E) = \{p\}$ and E is p-power torsion.*

Proof. Since $R/p \subset E$ is an essential extension, for any other associated prime $q = (0 :_R e)$ for $e \in R$, $re \in R/p$ for some r. Since p, q are prime and $r \notin q$ we have:

$$p = (0 :_R re) = ((- :_r e) :_R r) = (q :_R r) = q.$$

Thus $\mathrm{Ass}(E) = \{p\}$. For any annihilator I of an element in E, $R/I \subset E$ and thus $\mathrm{Ass}(R/I) = \{p\}$. Thus I is p-primary and $\sqrt{I} = p$.

□

Lemma 1.3. $E = E_R(k)$. *Then* $\mathrm{Hom}_R(k, E) = k$.

Proof. $\mathrm{Hom}_R(k, E)$ is a k-vector space and thus contains k as a direct summand; since $k \subset E$ is essential, we must obtain $\mathrm{Hom}_R(k, E) = k$ (E can't be m-torsion if $E \neq k$, since otherwise E is a k-vector space with a submodule independent of k). □

The following results concern the decomposition of an injective module.

Theorem 1.4. *Let E be an injective R-module, and let M be an R-module.*

(a) *There exists a decomposition*

$$E \cong \bigoplus_{p \in \operatorname{Spec} R} E_R(R/p)^{\mu_p}$$

where μ_p's are unique.

(b) *Let $E^\bullet$ be an injective resolution of M. For each E^i, (a) gives a decomposition*

$$E^i \cong \bigoplus_{p \in \operatorname{Spec} R} E_R(R/p)^{\mu_i(p,M)}.$$

*The number $\mu_i(p, M)$ depends only on p and M and is called the i-**th Bass number** of M with respect to p.*

Next comes a theorem frequently used later, which characterizes maps from a different ring to Matlis dual:

Theorem 1.5. *Let $E = E_R(k)$ and $\phi : (R, m, k) \to (S, n, l)$ be a finite, local homomorphism of local rings. Then $S^\vee = \operatorname{Hom}_R(S, E) = E_S(l)$.*

Proof. Finiteness gives $S^\vee$ is injective (by adjunction) and n-torsion. Then by injective decomposition 1.4 and 1.2 we are left to the only n-torsion component $S^\vee \cong E_S(l)^\mu$ for some μ.

Consider the isomorphisms below, induced by adjunctions and torsions:

$$\operatorname{Hom}_S(l, S^\vee) \cong \operatorname{Hom}_R(l, E) \cong \operatorname{Hom}_k(l, \operatorname{Hom}_R(k, E)) \cong \operatorname{Hom}_k(l, k) \cong l.$$

Also, we know $\operatorname{Hom}_S(l, E_S(l)) = l$ and we have $l^\mu = l$, so $\mu = 1$.

□

2 Matlis Duality

2.1 General duality

Proposition 2.1. *Let E be any injective module. The endofunctor $\mathrm{Hom}_R(-, E)$ on $\mathsf{Mod}(R)$ restricts to functors $\mathsf{Flat}(R) \longleftrightarrow \mathsf{Inj}(R)$.*

Lemma 2.2 ("taking tensors into hom")**.** *For modules, the natural map*

$$M \otimes_R \mathrm{Hom}_R(N, E) \to \mathrm{Hom}_R(\mathrm{Hom}_R(M, N), E)$$

becomes an isomorphism when M is finitely presented and E is injective.

Proof. Given a presentation $F \to G \to M \to 0$ of M, apply $-\otimes \mathrm{Hom}_R(N, E)$ and $\mathrm{Hom}_R(\mathrm{Hom}_R(-, N), E)$ to the sequence to obtain a "five-lemma" diagram (both functors are exact since E is injective).

Clearly it holds for finite rank free modules, thus the result.

□

Proof of Proposition 2.1. For flat F, $\mathrm{Hom}_R(F, E)$ is injective by adjunction (i.e. its Hom is exact).

If I is injective, by the lemma we have:

$$- \otimes \mathrm{Hom}_R(I, E) \cong \mathrm{Hom}_R(\mathrm{Hom}_R(-, I), E)$$

is exact, and thus $\mathrm{Tor}^1_R(-, \mathrm{Hom}_R(I, E)) = 0$ on finitely generated modules. But every module is a colimit of its finitely generated submodules and Tor commutes with colimits, so this holds for all R-modules. Thus $\mathrm{Hom}_R(I, E)$ is flat.

□

2.2 Matlis dual

Assume (R, m, k) is noetherian local. Let $E := E_R(k)$ be the injective hull of k.

Definition 2.3. Let the **Matlis dual** functor is the endofunctor $(-)^\vee := \mathrm{Hom}_R(-, E)$ on $\mathsf{Mod}(R)$.

E is actually a cogenerator for $\mathsf{Mod}(R)$:

Lemma 2.4. *A module M is zero if and only if $M^\vee$ is zero.*

Proof. For nonzero M, its cyclic submodule projects onto k: $R/I \to k \to E$, which only uses $k \hookrightarrow E$. This gives an nonzero element in $(R/I)^\vee$.

E being injective gives $M^\vee \to (R/I)^\vee$ surjective, so $M^\vee$ is nonzero.

□

Remark. By 1.5, it actually suffices to consider complete noetherian local ring R when it comes to Matlis dual, which is what we will see in Matlis Duality. Cohen Structure Theorem tells us that any notherian local ring $R \cong Q/I$ for complete Q, and by adjunction taking the Matlis dual under the two different basis is exactly the same.

Theorem 2.5 (Matlis duality)**.** *Let (R, m, k) be a complete noetherian local ring, $E := E_R(k)$. The Matlis dual functor $(-)^\vee$ restricts to equivalences, with inverse being itself, between the categories*

$$\mathsf{mod}(R) \overset{\cong}{\longleftrightarrow} \mathsf{art}(R)$$

which further restricts to

$$\mathsf{fl}(R) \overset{\cong}{\longleftrightarrow} \mathsf{fl}(R).$$

Remark. In particular, we have $M \cong M^{\vee\vee}$ when M is artinian or finitely generated, and M is finitely generated if and only if $M^\vee$ is artinian.

To prove the result, we need the following lemmas:

Lemma 2.6. *Note: (a) deals with a special case; (b) and (c) prove that $E^\vee = \widehat{R}$; (d) is a crucial technique on artinian modules.*

(a) *$N \cong N^{\vee\vee}$ for finite length N. In particular, if R is artinian, $E^\vee = R^{\vee\vee} = R$.*

(b) *E is an $\widehat{R}$-module and $E \cong E_{\widehat{R}}(k)$.*

(c) *$E^\vee = \widehat{R}$.*

(d) *R is artinian if and only if it is a submodule of $E^{\oplus n}$ for $n > 0$, where the smallest such n is $\dim_k(0 :_M m)$.*

Proof. Proof of (a): 1.3 gives $k^\vee = k$. Let (N_m) denotes the finite length series of N. By induction there is a short exact sequence:

$$0 \longrightarrow N_m \longrightarrow N_{m+1} \longrightarrow k \longrightarrow 0$$

Since $(-)^{\vee\vee}$ is exact, taking $(-)^{\vee\vee}$ we obtain a "five-lemma" diagram, and by induction the result holds.

Proof of (b): Say E is m^n-torsion, then for any $\hat{r} \in \widehat{R}$, $\hat{r}e$ is completely determined by the image of $\hat{r}$ in R/m^n, making E an $\widehat{R}$-module.

By 1.5, we also have $E = \bigcup_i (0 :_E m^i) = \operatorname{colim} \operatorname{Hom}_R(R/m^i, E) = \operatorname{colim} E_{R/m^i}(k)$. Since $R/m^i \cong \widehat{R}/m^i\widehat{R}$, going through exactly the same isomorphisms for $E_{\widehat{R}}(k)$ we obtain $E = E_{\widehat{R}}(k)$.

Proof of (c): By the above we obtain these isomorphisms:

$$\operatorname{Hom}_R(E, E) \cong \operatorname{Hom}_R(\operatorname{colim} E_{R/m^i}(k), E) \cong \lim \operatorname{Hom}_R(E_{R/m^i}(k), E).$$

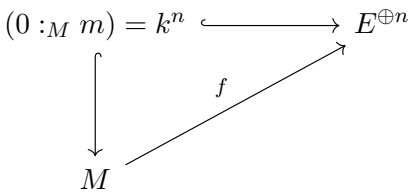
We have already showed that Matlis dual over R and R/m^i are the same, which translates exactly into $\operatorname{Hom}_R(E_{R/m^i}(k), E) \cong \operatorname{Hom}_{R/m^i}(E_{R/m^i}(k), E_{R/m^i}(k))$, which is R/m^i by part (a). Hence:

$$E^\vee = \lim R/m^i = \widehat{R}.$$

Proof of (d): One direction is done after we prove E is artinian below.

If M is artinian, then $(0 :_M m)$ is essential in M: for any submodule N of M, since it is artinian it must contain a simple submodule $L \cong k$, and thus $L \subset (0 :_M m)$.

$(0 :_M m) = k^n$ as a k-vector space. Consider the following two modules that $(0 :_M m)$ embeds into:



Clearly $\ker(f) \cap k^n = 0$, and since $k^n \subset M$ is essential, $\ker(f) = 0$.

Such n is clearly the smallest since k^n embeds into any possible E^i.

□

The proof of Matlis duality comes in three steps: prove E is artinian (also needed for 2.6(d)), $(-)^\vee$ does land in $\mathsf{mod}(R)$ and $\mathsf{art}(R)$, and it is an equivalence in both cases.

proof of theorem 2.5. The steps are:

Step 1: For any descending chain of submodules in E. $E = E_0 \supset E_1 \supset E_2 \supset \dots$, taking $(-)^\vee$ and by 2.6(c) we have surjections $R \to E_1^\vee \to E_2^\vee \to \dots$, and $K_i := \ker(R \to E_i^\vee)$ form an ascending chain of ideals and must stabilize.

Then $K_i = K_{i+1}$ implies $\ker(E_i \to E_{i+1}) = (E_{i+1}/E_i)^\vee = 0$, and by 2.4, E is artinian.

Step 2: First assume M is finitely generated. By 2.2, $M^{\vee\vee} \cong M \otimes \mathrm{Hom}_R(E, E) = M \otimes E^\vee = M$ since $E^\vee = \widehat{R} = R$.

Now assume M is artinian, then by 2.6(d) and E being artinian, we obtain a left exact sequence, applying $(-)^\vee$ to which we have:

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \longrightarrow & E^n & \longrightarrow & E^m \\ \downarrow & & \downarrow & & \downarrow\cong & & \downarrow\cong \\ 0 & \longrightarrow & M^{\vee\vee} & \longrightarrow & (E^{\vee\vee})^n & \longrightarrow & (E^{\vee\vee})^m \end{array} .$$

Step 3: This is a delicate use of 2.6(d). Characterization of $\mathsf{mod}(R)$ and $\mathsf{art}(R)$, $R^n \twoheadrightarrow M$ and $M^\vee \hookrightarrow E^n$, dualize to each other.

□

3 Derived category

3.1 Basic construction

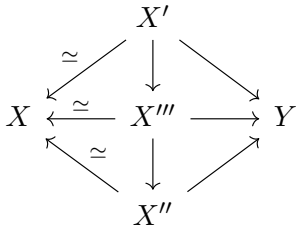
Let $\mathsf{C}(R)$ denote the abelian category of chain complexes. Note that the morphisms are the Hom complex below at degree 0, so they do form an abelian group.

Definition 3.1 (derived category)**.** The **derived category** $\mathsf{D}(R)$ is the localization of the category $\mathsf{C}(R)$ with respect to quasi-isomorphisms i.e. for any $\mathsf{C}(R) \to \mathsf{C}'$ sending quasi-isomorphisms to isomorphisms, it factors uniquely through $\mathsf{D}(R)$.

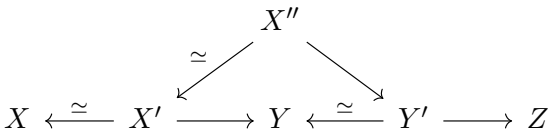
$\mathsf{D}(R)$ is a *triangulated category*, with the following ingredients:

(a) The objects of $\mathsf{D}(R)$ are the same as $\mathsf{C}(R)$.

(b) To demonstrate quasi-isomorphic objects in $\mathsf{D}(R)$ are isomorphic, the morphisms $X \to Y$ of $\mathsf{D}(R)$ are in the form $X \xleftarrow{\simeq} X' \to Y$, and two morphisms are equivalent if there is X''' to complete the roof:



(c) Composition of morphism is given by a semiprojective resolution: for $X \xleftarrow{\simeq} X' \to Y$ and $Y \xleftarrow{\simeq} Y' \to Z$, their composition is $X \xleftarrow{\simeq} X'' \to Z$ such that:



Here X'' can be chosen to be the semiprojective resolution $P \xrightarrow{\simeq} X$, ensured by 3.6. Semiprojective complexes preserve quasi-isomorphisms, so $H_0(\mathrm{Hom}(P, Y')) = H_0(\mathrm{Hom}(P, Y))$, and we can find the lift $X'' = P \to Y'$.

(d) The suspension functor is given by Σ.

(e) The exact triangles are isomorphic to $X \xrightarrow{f} Y \to \mathrm{cone}(f) \to$, where $\mathrm{cone}(f)$ in $\mathsf{D}(R)$ is defined as $\mathrm{cone}(f')$, where $f = X \xleftarrow{\simeq} X' \xrightarrow{f'} Y$ (but we will always assume we have chosen a representative $X \xrightarrow{f'} Y$). $\mathrm{cone}(f)$ is unique up to noncanonical isomorphisms.

Moreover, note that every short exact sequence $0 \to X \to Y \to Z \to 0$ becomes an exact triangle $X \to Y \to Z \to$ in $\mathsf{D}(R)$, where it can be proved that Z is quasi-isomorphic to $\mathrm{cone}(X \to Y)$.

(f) $H : \mathsf{D}(R) \to \mathsf{Mod}(R)$ is a well-defined functor since it works well with quasi-isomorphisms. It is a cohomological functor and thus induces a long exact sequence for any exact triangles.

(g) Using (f) in $\mathsf{C}(R)$, we can show α is quasi-isomorphism if and only if $\mathrm{cone}(\alpha)$ is acyclic.

3.2 Important objects and functors in $\mathsf{C}(R)$

The overarching property of $\mathsf{D}(R)$ is that "homotopy equivalent" objects are viewed identical, so that we can define many homotopy invariant objects and functors. To define these, we need some facts in $\mathsf{C}(R)$.

Definition 3.2 (Hom and tensor)**.** For any complex X, Y, we define their **Hom complex** and **tensor complex** as usual.

Definition 3.3. Consider complexes in $\mathsf{C}(R)$.

(a) P is **semiprojective** if $\mathrm{Hom}_R(P, -)$ is exact and preserves quasi-isomorphism.

(b) I is **semiinjective** if $\mathrm{Hom}_R(-, I)$ is exact and preserves quasi-isomorphism.

(c) S is **semiflat** if $F \otimes_R -$ is exact and preserves quasi-isomorphism.

(d) F is **semifree** if it admits a filtration by subcomplexes

$$0 \subset F(0) \subset F(1) \subset \cdots \subset F$$

where $\operatorname{colim} F(i) = F$ and $F(i)/F(i-1)$ is a sum of shifts of R (graded free with *zero differential*).

Note that F is more than each degree being free module when it is not bounded below: the definition requires differentials of elements in stage i lie in previous stages. Equivalently, semifree F admits a basis having this property.

(e) J is **semicofree** if it admits a filtration by subcomplexes

$$J = J(0) \supset J(1) \supset \dots$$

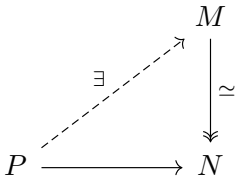
such that $\varprojlim J/J(i) \cong J$ and $J(i)/J(i+1) \cong (F^i)^\vee$, where F^i is a semifree complex and $(-)^\vee = \mathrm{Hom}_R(-, E)$ for E being the *injective cogenerator* of $\mathsf{Mod}(R)$, $\mathrm{Hom}_{\mathbb{Z}}(R, \mathbb{Q}/\mathbb{Z})$.

Proposition 3.4 (Semiprojectivity)**.** *The following are equivalent:*

(a) *P is semiprojective,*

(b) *P_i is projective and preserves acyclicity.*

(c) *For any diagram*



there exists a lift, unique up to homotopy.

Remark. There are analogous statements for semiinjective and semiflat complexes.

Proof. (a) $\Leftrightarrow$ (b) is straight forwards, while equivalence to (c) requires some constructions.

If $\mathrm{Hom}_R(P, -)$ is exact, apply it to the exact sequence $0 \to S^i(A) \to S^i(B) \to S^i(C) \to 0$ with $A \to B \to C$ in the i-th degree, by looking at degree 0 we know P_i is projective; if P_i is projective, then $\mathrm{Hom}_R(P_i, -)$ and $\prod$ are both exact in $\mathsf{Mod}(R)$, and thus $\mathrm{Hom}_R(P, -)$ is exact.

Preserving $\simeq$ $\Leftrightarrow$ preserving acyclic complexes is a direct consequence of 3.3(g) and that cone commutes with Hom. $\square$

Proposition 3.5. *Semifree complexes are semiprojective; semiprojective complexes are semiflat; semicofree complexes are semiinjective. Bounded below complexes of projective/flat modules are semiprojective/semiflat; bounded above complexes of injective modules are semiinjective.*

Remark. We will need a classical characterization on inverse limits: given an inverse system of R-complexes (X^i) with maps (δ^i), there is an exact sequence

$$0 \longrightarrow \varprojlim X^i \longrightarrow \prod_{i\geq 0} X^i \xrightarrow{d^X} \prod_{i\geq 0} X^i$$

where $d^X : (x^i) \mapsto (x^i - \delta^{i+1}(x^{i+1}))$.

It is well known that if the system is surjective, then d^X is surjective, and we obtain a short exact sequence. If furthermore each X^i is acyclic, then since $\prod$ is exact $\prod X^i$ is also acyclic, and by the long exact sequence, $\varprojlim X^i$ must be acyclic.

Proof. First we prove that a semifree complexes F is semiprojective. By 3.4, it suffices to show $\mathrm{Hom}_R(F, A) \simeq 0$ for $A \simeq 0$. To begin with, we use induction to prove it for $F(i)$. Consider the short exact sequence of complexes

$$0 \longrightarrow F(i-1) \longrightarrow F(i) \longrightarrow F(i)/F(i-1) \longrightarrow 0 \ .$$

The sequence is split as a graded module since $F(i)/F(i-1)$ is projective in each degree. Also, $\mathrm{Hom}_R(F(i)/F(i-1), A) \cong A^m \simeq 0$. Thus $\mathrm{Hom}_R(F(i), A) \simeq 0$ as the direct sum of two acyclic complexes.

Then note that, since $\mathrm{Hom}_R(F(i), A) = \mathrm{Hom}_R(F(i-1), A) \oplus A^{m_i}$ for some m_i, each map in the following system is a projection:

$$\cdots \longrightarrow \mathrm{Hom}_R(F(2), A) \longrightarrow \mathrm{Hom}_R(F(1), A) \longrightarrow \mathrm{Hom}_R(F(0), A) \longrightarrow 0$$

Then by the previous remark, since each term is acyclic, $\mathrm{Hom}_R(F, A) = \mathrm{Hom}_R(\mathrm{colim}\, F(i), A) = \varprojlim \mathrm{Hom}(F(i), A) \simeq 0$.

The proofs for semiflat and semiinjective modules are similar.

To prove a bounded below complex P of projective modules is semiprojective, let $P(i) = P_i \to P_{i-1} \to \cdots \to P_0 \to 0$, and the rest is exactly the same: induction shows $\mathrm{Hom}(P(i), -)$ perserves acyclicity since $P(i)/P(i-1)$ being projective implies splitting; then surjective inverse system shows $\mathrm{Hom}(P, -)$ perserves acyclicity.

The bounded above case is similar.

□

Theorem 3.6. *Every R-complex M admits a semiprojective (semifree) resolution $P \xrightarrow{\simeq} M$ and a smemiinjective (semicofree) resolution $M \xhookrightarrow{\simeq} I$. They are unique up to homotopy equivalence.*

Proof. We omit the proof of the existence part. The proof for semifree resolution roughly goes as follows. For the base case, we fix $Z(0)$ to be a generating set of $Z(M)$, $F(0) = \bigoplus_{z\in Z(0)} \Sigma^{|z|} R$, and canonical map $\varepsilon(0) : F(0) \to M$.

The induction step comes in two parts: we first fix $Z(i)$ whose homology class is the generating set for $\ker(\mathrm{H}(\varepsilon(i))$; these are the "new variables" we want to attach. Then we define $F(i+1)$ to be the cone of the natural map from $\bigoplus_{z\in Z(i)} \Sigma^{|z|} R \to F(i)$; this is "attaching the new variables". It remains to check the compatibility of this construction.

The limit stage defines $F := \operatorname{colim} F(i)$, $\varepsilon := \operatorname{colim} \varepsilon(i)$. The above arguments ensure that F is semifree, $\ker(\mathrm{H}(\varepsilon)) = 0$ (ε is quasi-isomorphism), and ε is surjective.

The uniqueness part follows immediately from theorems like 3.4(c).

□

Following this theorem, we can take better care of an important subcategory of $\mathsf{C}(R)$.

Lemma 3.7. *Let R be Noetherian. For any complex M, $\mathrm{H}(M)$ is finitely generated if and only if M is quasi-isomorphic to a bounded complex of finitely generated modules. We denote the full subcategory of all such complexes as $\mathsf{C}^b(R)$, and we denote the same full subcategory of $\mathsf{D}(R)$ as $\mathsf{D}^b(R)$.*

Proof. Since R is Noetherian, if M_i is finitely generated, then it is Noetherian and any submodule of M_i is also finitely generated, thereby $\mathrm{H}(M)$ is finitely generated.

If $\mathrm{H}(M)$ is finitely generated, then by the constructions in 3.6 we can choose a semifree resolution $P \xrightarrow{\sim} M$ such that it is bounded below (the degree of the variables is bounded below) and finitely generated in each degree (finite number of variables each time). Let $m = \sup\{i : \mathrm{H}_i(M) \neq 0\}$, then truncating P at m we obtain a bounded complex of finitely generated modules that is quasi-isomorphic to M.

□

3.3 Constructions in $\mathsf{D}(R)$

We are finally prepared to introduce two crucial functors in $\mathsf{D}(R)$.

Definition 3.8. For each R-complex M, $\mathrm{RHom}_R(M, -) := \mathrm{Hom}_R(P, -)$, $M \otimes^{\mathrm{L}}_R - := P \otimes_R -$, where $P \xrightarrow{\simeq} M$ is a semiprojective resolution of M.

We need the follwing remarks:

(a) The functors are independent of resolutions since semiprojective resolutions are homotopy equivalent, and both Hom_R and $\otimes_R$ preserve homotopy equivalence (so the results are quasi-isomorphic); also, semiprojectivity and semiflatness of P makes sure the functors are well-defined in $\mathsf{D}(R)$.

(b) $\mathrm{RHom}_R(M, N)$ and $M \otimes^{\mathrm{L}}_R N$ can also be computed by replacing N by its semiinjective resolution and its semiflat resolution, respectively.

(c) Define $\mathrm{Tor}^R(M, N) := \mathrm{H}(M \otimes^{\mathrm{L}}_R N)$ and $\mathrm{Ext}_R(M, N) := \mathrm{H}(\mathrm{RHom}_R(M, N))$.

Definition 3.9. A full subcategory S of a triangulated category T is **thick** if it is a triangulated subcategory (closed under suspension and cone) and is closed under direct summands.

For an object X of T, let $\mathsf{thick}(X)$ denote the **thick closure** of X, which is the smallest thick subcategory of T containing X (well-defined since intersections of thick subcategories are thick).

Remark. The *eventual vanishing of a homological functor* can usually produce a thick subcategory. For example, given R-complex M, the full subcategory $\mathsf{D}^b(R)$ of N with $\mathrm{H}(N)$ finitely generated (although this requires more than eventual vanishing), the full subcategory of N such that $\mathrm{Tor}^R_{\gg 0}(M, N) = 0$, and the full subcategory of N such that $\mathrm{Ext}^{\gg 0}_R(M, N) = 0$ are all thick subcategory.

There is no doubt of closure under suspension and direct summand. For any exact triangle $X \to Y \to Z \to$, homological functor F ensures $F(X) \to F(Y) \to F(Z) \to$ is also an exact triangle, producing a long exaxct sequence

$$F_n(X) \xrightarrow{\alpha} F_n(Y) \longrightarrow F_n(Z) \longrightarrow F_{n-1}(X) \xrightarrow{\beta} F_{n-1}(Y) \longrightarrow \cdots.$$

Therefore, if X and Y has eventual vanishing F, so does Z. Moreover, if $\mathrm{H}_n(Y)$ and $\mathrm{H}_{n-1}(X)$ are finitely generated, then $0 \to \operatorname{coker}(\alpha) \to \mathrm{H}_n(Z) \to \ker(\beta) \to 0$ is an exact sequence with the two sides finitely generated. Thus $\mathrm{H}_n(Z)$ is also finitely generated.

Proposition 3.10 (Perfect subcategory)**.** *Let R be a Noetherian ring and let $\mathsf{Perf}(R)$ be the full subcategory of $\mathsf{D}(R)$ of those isomorphic to a bounded complex of finitely generated projective R-modules.*

Then $\mathsf{Perf}(R)$ is thick in $\mathsf{D}^b(R)$ and thus $\mathsf{D}(R)$, and $\mathsf{Perf}(R) = \mathsf{thick}(R)$.

Proof. $\mathsf{Perf}(R)$ is thick since suspension, $\oplus$, and direct summand all preserve projectivity in each degree.

By definition $\mathsf{thick}(R) \subset \mathsf{Perf}(R)$. Now consider the exact triangle $P^{n-1} \to P^n \to \Sigma^n P_n \to$, where $P^i = (0 \to P_i \to P_{i-1} \to \cdots \to 0)$. $\Sigma^n P_n$ is in $\mathsf{thick}(R)$ since free modules and their summands are, and by induction P_{n-1} belongs to $\mathsf{thick}(R)$. Rotating the triangle gives the result.

□

Definition 3.11 (Homotopy colimit)**.** Given a sequence in $\mathsf{D}(R)$

$$X^0 \xrightarrow{\alpha^0} X^1 \xrightarrow{\alpha^1} X^2 \longrightarrow \dots$$

a **homotopy colimit** of the sequence, $\operatorname{hocolim}(X^i)$, fits in the exact triangle

$$\bigoplus X^i \xrightarrow{1-\alpha} \bigoplus X^i \longrightarrow \operatorname{hocolim}(X^i) \longrightarrow$$

where $1 - \alpha : x^i e_i \mapsto (x^i e_i - \alpha^i(x^i) e_{i+1})$.

As cone is a model-free version of coker, homotopy colimit is the homotopy-invariant version of colimit; once a chain-level model is nailed down, homotopy colimits just become usual colimits. For example, if we choose $M \xrightarrow{\simeq} J$, then $\operatorname{hocolim} \mathrm{RHom}(A_n, M) \simeq \operatorname{colim} \mathrm{Hom}(A_n, J)$.

Lemma 3.12. $\mathrm{H}(\operatorname{hocolim} X^i) \cong \operatorname{colim} \mathrm{H}(X^i)$ *as an isomorphism of graded R-modules.*

Proof. This demonstrates the ideas above. Applying H to the exact triangle, we obtain

$$\bigoplus \mathrm{H}(X^i) \xrightarrow{1-\mathrm{H}(\alpha)} \bigoplus \mathrm{H}(X^i) \longrightarrow \mathrm{H}(\operatorname{hocolim}(X^i)) \longrightarrow$$

Since $1 - \mathrm{H}(\alpha)$ is now a concrete map of R-modules (same for R-complexes), it must be injective, and the triangle becomes a short exact sequence where $\operatorname{colim} \mathrm{H}(X^i) = \operatorname{coker}(1 - \mathrm{H}(\alpha)) \cong \mathrm{H}(\operatorname{hocolim}(X^i))$.

□

Remark (How to produce an exact functor on $\mathsf{D}(R)$). Let F be an additive functor on $\mathsf{Mod}(R)$, and extend it degreewise to $F : \mathsf{C}(R) \to \mathsf{C}(R)$, which is also additive. We notice the following facts.

(a) Since F is additive, F commutes with suspension and cone, and it preserves homotopy equivalence. Thus it is a well-defined exact triangulated functor on the homotopy category $\mathsf{K}(R)$.

(b) By replacing the inputs with resolutions, we obtain a well-defined exact functor $\mathrm{R}F : \mathsf{D}(R) \to \mathsf{D}(R)$.

(c) If additionally F is left exact, then we may recover F in the lowest degree. For instance, given a complex M with no negative degree, we can show that $\mathrm{H}_0(\mathrm{R}F(M)) \simeq F(\mathrm{H}_0(M))$. The same goes for right exact functors on $\mathsf{Mod}(R)$.

4 Local cohomology

4.1 I-power torsion

Definition 4.1 (Local cohomology)**.** For an R-module M, its **I-power torsion submodule** is

$$\Gamma_I(M) := \bigcup_{n\geq 0}(0 :_M I^n) = \operatorname{colim}_n \operatorname{Hom}_R(R/I^n, M).$$

By the remark at the end of the last section, this defines a functor $\mathrm{R}\Gamma_I : \mathsf{D}(R) \to \mathsf{D}(R)$ such that $\mathrm{R}\Gamma_I(M) := \Gamma_I(J)$, where J is a semiinjective resolution of M.

Note that Γ_I is left exact on $\mathsf{Mod}(R)$ since $\operatorname{Hom}_R(R/I^n, -)$ is left exact and the filtered colimit is exact.

For R-complex M, its **local cohomology supported on** I is

$$\mathrm{H}_I(M) := \mathrm{H}(\mathrm{R}\Gamma_I(M)).$$

Note that we always assume R is Noetherian, and $\sqrt{I} = \sqrt{I'}$ implies $\mathrm{H}_I(M) \cong \mathrm{H}_{I'}(M)$ since $I^n \subset I'$ for some n.

Remark. $\mathrm{R}\Gamma_I(M)$ is after all the I-power torsion of the semiinjective resolution J of M at every degree, so naturally: there is $\mathrm{R}\Gamma_I(M) \to M$, and for any $M \to N$ in $\mathsf{D}(R)$ there is induced $\mathrm{R}\Gamma_I(M) \to \mathrm{R}\Gamma_I(N)$.

Moreover, by 4.8 if M has I-power torsion homology, then $M \to N$ factors as $M \to \mathrm{R}\Gamma_I(N) \to N$ (since $M \simeq \mathrm{R}\Gamma_I(M)$ in this case).

As Γ_I is defined using colim and Hom in $\mathsf{Mod}(R)$, we can define $\mathrm{R}\Gamma_I$ using $\operatorname{hocolim}$ and RHom in $\mathsf{D}(R)$.

Proposition 4.2. *The natural map* $\operatorname{hocolim} \operatorname{RHom}_R(R/I^n, M) \simeq \mathrm{R}\Gamma_I(M)$ *is an isomorphism in* $\mathsf{D}(R)$*. In particular,* $\mathrm{H}_I(M) = \mathrm{H}(\mathrm{R}\Gamma_I(M)) \cong \mathrm{H}(\operatorname{hocolim} \operatorname{RHom}_R(R/I^n, M)) \cong \operatorname{colim} \operatorname{Ext}_R(R/I^n, M)$.

Proof. Choose any semiinjective resolution J of M. Then $\operatorname{hocolim} \operatorname{RHom}_R(R/I^n, M) \simeq \operatorname{colim} \operatorname{Hom}_R(R/I^n, J) = \Gamma_I(J) = \mathrm{R}\Gamma_I(M)$.

□

4.2 Stable Koszul complex

There is a natural map of complexes from the Koszul complex $K(\mathbf{x})$ on $\mathbf{x}$ to R/I, where $\mathbf{x} = (x_1, \ldots, x_c)$ is a generating set of I. Note that $\operatorname{colim} \operatorname{Hom}_R(R/I^n, -) = \operatorname{colim} \operatorname{Hom}_R(R/I_n, -)$, where $I_n = (\mathbf{x}^n) := (x_1^n, \ldots, x_c^n)$. This leads to the possibility of computing local cohomology using some sort of Koszul complexes.

Definition 4.3. The **stable Koszul complex on** $\mathbf{x}$ is $K_\infty(\mathbf{x}) := \operatorname{colim}_n \operatorname{Hom}_R(K(\mathbf{x}^n), R)$.

We need a useful lemma:

Lemma 4.4. *For* P *in* $\mathsf{Perf}(R)$*, we have*

$$\operatorname{Hom}_R(P, R) \otimes_R M \cong \operatorname{Hom}_R(P, M)$$

for any complex M*.*

Dually, $\operatorname{Hom}_R(R, J) \otimes_R M \cong \operatorname{Hom}_R(M, J)$ *for* J *a bounded complex of finitely generated injective modules.*

Proposition 4.5. *Let I and I_n be defined as above.*

(a) *There is a natural map* $\mathrm{R}\Gamma_I(M) \to K_\infty(\mathbf{x}) \otimes_R M$.

(b) $K_\infty(\mathbf{x}) \cong \bigotimes_{i=1}^c K_\infty(x_i)$.

(c) $K_\infty(\mathbf{x})$ *is in the form of:*

$$K_\infty(\mathbf{x}) = 0 \to R \to \bigoplus_{i=1}^c R_{x_i} \to \bigoplus_{i<j} R_{x_i x_j} \to \cdots \to R_{x_1 x_2 \ldots x_c} \to 0$$

Proof. Proof of (a): Consider the following maps:

$$\mathrm{R}\Gamma_I(M) \simeq \operatorname{hocolim} \mathrm{RHom}_R(R/I_n, M) \to \operatorname{hocolim} \mathrm{RHom}_R(K(\mathbf{x}^n), M) \simeq \operatorname{colim} \mathrm{Hom}_R(K(\mathbf{x}^n), R) \otimes_R M.$$

The middle morphism exists by triangulated axioms ($\operatorname{hocolim}$ is essentially a cone), and the last map follows from 4.4 since $K(\mathbf{x}^n)$ is a bounded complex of free modules.

Proof of (b): The KEY is that Koszul complexes are self-dual as chain complexes:

$$\mathrm{Hom}_R(K(\mathbf{x}^n), R) \cong \Sigma^{-c} K(\mathbf{x}^n)$$

which can be proved by induction (tensor construction of Koszul complex). Using this we can write the dual Koszul complex as individual pieces:

$$\mathrm{Hom}_R(K(\mathbf{x}^n), R) \cong \Sigma^{-c} K(\mathbf{x}^n) = \bigotimes_{i=1}^c \Sigma^{-1} K(x_i^n) \cong \bigotimes_{i=1}^c \mathrm{Hom}_R(K(x_i^n), R).$$

After checking compatibility using diagrams, we can apply colim both sides, and since tensors (left adjoint) commute with colimits, we have $K_\infty(\mathbf{x}) \cong \bigotimes_{i=1}^c K_\infty(x_i)$.

Proof of (c): Observe that for single element x, $K_\infty(x)$ is the colimit of the diagram

$$\begin{array}{ccccccc} R & = & R & = & R & = & \cdots \\ \downarrow{\scriptstyle x} & & \downarrow{\scriptstyle x^2} & & \downarrow{\scriptstyle x^3} & & \\ R & \xrightarrow{x} & R & \xrightarrow{x} & R & \xrightarrow{x} & \cdots \end{array}$$

where the columns are individual $K(x^n)$. The bottom row is isomorphic to R_x, with the isomorphism given by $a \mapsto \frac{a}{x^n}$ for a in the n-th copy (then it is compatible with the maps: $\frac{a}{x^n} = \frac{xa}{x^{n+1}}$). Thus:

$$K_\infty(x) = 0 \to R \to R_x \to 0.$$

for R in degree 0.

Note that $R_x \otimes_R R_y \cong (R_x)_y \cong R_{xy}$, and (c) follows from (b) and the above.

□

The map turns out to be a quasi-isomorphism:

Theorem 4.6. *For any complex M,* $\mathrm{R}\Gamma_I(M) \to K_\infty(\mathbf{x}) \otimes_R M$ *is an isomorphism in* $\mathsf{D}(R)$.

4.3 Properties of $\mathrm{R}\Gamma_I$

4.6 gives us a powerful tool to prove many facts on local cohomology.

Proposition 4.7. *Let M be an R-complex.*

(a) $\mathrm{H}_I(-)$ *commutes with direct sums.*

(b) *If M is an R-module, $\mathrm{H}^i_I(M) = 0$ for $i > \operatorname{ara}(I) := \inf\{r : \exists\ a_1, \ldots, a_r$ such that $I = \sqrt{(a_1, \ldots, a_n)}\}$.*

In particular, if (R, m) is local and $I = m$, then the conclusion holds for $i > \operatorname{ara}(m) = \dim(R)$ (Local ring has a system of parameters which cannot be longer than $\dim(R)$ by Krull's height).

(c) *If $R \to S$ is a flat ring map, then $\mathrm{H}_I(M) \otimes_R S \cong \mathrm{H}_I(M \otimes_R S) \cong \mathrm{H}_{IS}(M \otimes_R S)$.*

In particular, for any mutiplicatively closed set U, letting $S = U^{-1}R$ we get H_I commutes with localization and is stable under localization of I.

Proof. Proof of (a): Use 4.6 and the fact that tensor and H commutes with direct sums.

Proof of (b): Use 4.6 to see $\mathrm{R}\Gamma_I(M)$ is supported from $-\operatorname{ara}(I)$ to 0 since $\mathrm{H}_I(-) = \mathrm{H}_{\sqrt{I}}(-)$ and $K_\infty(\mathbf{x})$ has the same length of $\mathbf{x}$. Use cohomological convention to get the result.

Proof of (c): The first isomorphism comes from S being a flat R-module: S commutes with the homology and 4.6 gives the result.

The second uses 4.6 and the fact that localization is stable under base change.

□

The next theorem says derived I-power torsion can be detected by the homology of M.

Theorem 4.8. *The natural map $\mathrm{R}\Gamma_I(M) \to M$ is an isomorphism in $\mathsf{D}(R)$ if and only if $\mathrm{H}(M)$ is I-power torsion.*

The proposition below tells us when $\mathrm{R}\Gamma_I$ commutes with RHom:

Proposition 4.9. *The natural map $\mathrm{RHom}_R(M, \mathrm{R}\Gamma_I(N)) \to \mathrm{R}\Gamma_I(\mathrm{RHom}_R(M, N))$ is an isomorphism in $\mathsf{D}(R)$ in the following cases:*

(a) *For any N when M is in $\mathsf{Perf}(R)$;*

(b) *When M is in $\mathsf{D}^b(R)$ and $\mathrm{H}_i(N) = 0$ for $i \gg 0$.*

Proof. Proof of (a): Consider the category of all L in $\mathsf{D}(R)$ such that

$$\mathrm{RHom}_R(L, \mathrm{R}\Gamma_I(N)) \to \mathrm{R}\Gamma_I(\mathrm{RHom}_R(L, N))$$

is an isomorphism. This is a thick subcategory since $\mathrm{R}\Gamma_I$ and RHom both are exact and commute with direct sums and suspensions. R is clearly in this category, so $\mathsf{Perf}(R) = \mathsf{thick}(R)$ is a subcategory of it.

Proof of (b): note that under the assumption M has a bounded below semiprojective resolution, and N has a bounded above semiinjective resolution. We can show that the isomorphism in $\mathsf{D}(R)$ holds for bounded complexes.

□

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